

CPSC 304 Project Cover Page

Milestone #: 3

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Group Number: 80

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Milestone 3 Project Check In

Topic: Urban Bike Infrastructure Maintenance System

Project Summary

Our project is about urban bike infrastructure maintenance and operations. It concerns the real-world processes involved in operating and maintaining a city-wide shared bicycle system, including ensuring that bicycles deployed across urban stations remain safe, functional, and available for public use. This domain involves coordination between physical assets, maintenance staff, and operational policies to support reliable and efficient urban transportation.

Timeline and Task Assignment

Note: this is subjective to change.

Week 9 (Milestone 3 due on Friday 10 pm)

Overall Task:

- By Friday at 10:00 PM on March 6, the team will complete the Milestone 3 deliverables, ensure that all project deliverables from Milestones 1 to 3 are uploaded to the repository, and confirm that all members can clone, pull, and push to the team repository successfully.

Boyi's Tasks:

- Upload the Milestone 2 deliverables to the repository.
- Add DROP TABLE statements in dependency order for all tables in db.sql
- Verify that the SQL script can reset the database without manual cleanup.

Runtan's Tasks:

- Create and organize the db.sql file.
- Add all CREATE TABLE statements from the Milestone 2 schema design into the script.
- Verify primary keys, foreign keys, and integrity constraints in Oracle.

Xinghao's Tasks:

- Set up the repository structure based on the sample project.
- Upload the Milestone 1 deliverables to the repository.
- Add at least five tuples for each table and test that the insertion script runs successfully.

Week 10 (Check-in with TA)

Overall Task:

- By Thursday (March 12), from 7:45 PM to 8:00 PM, each team member must attend the TA check-in meeting and demonstrate understanding of the project structure, database schema, and implementation plan for frontend, backend, and SQL queries.

Boyi's Tasks:

- Attend the Zoom meeting and demonstrate an understanding of the project structure.
- Be able to run the sample project and make commits to the repository.
- Review the course slides related to feature backend and frontend implementation.
- Discuss with the team to assign Non-Hardcoded Queries and Hardcoded Queries to each member.

Runtan's Tasks:

- Attend the Zoom meeting and demonstrate an understanding of the project structure.
- Be able to run the sample project and make commits to the repository.
- Review the course slides related to feature backend and frontend implementation.
- Discuss with the team to assign Non-Hardcoded Queries and Hardcoded Queries to each member.

Xinghao's Tasks:

- Attend the Zoom meeting and demonstrate an understanding of the project structure.
- Be able to run the sample project and make commits to the repository.
- Review the course slides related to feature backend and frontend implementation.
- Discuss with the team to assign Non-Hardcoded Queries and Hardcoded Queries to each member.

Week 11

Overall Task:

- Begin implementing database queries, backend APIs, and frontend forms. The team will complete the first working version of the application flow: frontend form → backend endpoint → SQL query → displayed result.

Boyi's Tasks:

- Implement 2 Non-Hardcoded Queries
- Implement 1 Hardcoded Query
- Connect the corresponding frontend forms to the backend logic.

Runtan's Tasks:

- Implement 2 Non-Hardcoded Queries
- Implement 1 Hardcoded Query
- Connect the corresponding frontend forms to the backend logic.

Xinghao's Tasks:

- Implement 3 Non-Hardcoded Queries
- Implement 1 Hardcoded Query

Week 12

Overall Task:

- Continue implementing advanced SQL queries and improve GUI functionality.

Boyi's Tasks:

- Continue and complete the implementation.
- Improve the display of query results in GUI.

Runtan's Tasks:

- Continue and complete the implementation.
- Improve error handling and user notifications.

Xinghao's Tasks:

- Continue and complete the implementation.
- Improve GUI usability and layout.
- Ensure user input fields function correctly

Week 13 (Milestone 4 due on Thursday; Milestone 6 due on Friday)

Overall Task:

- By Thursday at 10pm (April 2), the Milestone 4 deliverables must be completed. Finalize the project implementation and prepare all required materials.
- By Friday at 10pm (April 3), the Milestone 6 Individual and Peer Assessment must be completed.

Boyi's Tasks:

- Complete the Milestone 4 deliverables and update the README.md on GitHub
- Finalize the SQL script.
- Complete the Individual and Peer Assessment quiz.

Runtan's Tasks:

- Complete the Milestone 4 deliverables and update the README.md on GitHub
- Prepare documentation describing the final database schema.
- Complete the Individual and Peer Assessment quiz.

Xinghao's Tasks:

- Complete the Milestone 4 deliverables and update the README.md on GitHub
- Ensure the repository structure contains all required files.
- Complete the Individual and Peer Assessment quiz.

Week 14 (Milestone 5 due on Tuesday)

Overall Task:

- Prepare for the project demo and verify that all queries and GUI features work correctly.

Boyi's Tasks:

- Demonstrate the functionality of the assigned Non-Hardcoded and Hardcoded queries.
- Prepare to answer technical questions during the demo.

Runtan's Tasks:

- Demonstrate the functionality of the assigned Non-Hardcoded and Hardcoded queries.
- Prepare to answer technical questions during the demo.

Xinghao's Tasks:

- Demonstrate the functionality of the assigned Non-Hardcoded and Hardcoded queries.
- Prepare to answer technical questions during the demo

Planned Backend API Endpoints

The backend will provide several API endpoints so that the frontend can request data from the Oracle database. These endpoints will execute SQL queries and return results to the frontend for display.

Planned endpoints include:

- **GET /bikes**
Retrieve all bikes and their basic information.
- **GET /bikes/:bikeId**
Retrieve the information of a specific bike based on its BikeID.
- **GET /bikes/status/:status**
Retrieve bikes filtered by status (e.g., Active or Repair).
- **GET /maintenance/bike/:bikeId**
Retrieve maintenance records and tasks associated with a specific bike.
- **GET /issues/score/:score**
Retrieve issues with a given condition score.
- **GET /api/bikes?status=Repair**
Retrieve all bikes currently under repair (Hardcoded query).
- **GET /reports/component-usage**
Retrieve aggregated statistics about component usage (Hardcoded query).
- **GET /reports/technician-task-summary**
Retrieve the number of maintenance tasks assigned to each technician (Hardcoded query).
- **PUT /api/bikes/:bikeId**
Update the operational status of a bike
- **POST /api/maintenance**
Add new maintenance
- **POST /api/maintenances/:maintenanceld/tasks**
Add new maintenance task
- **GET /api/maintenances/:maintenanceld/tasks**
Get all the maintenance task from a specific maintenance
- **DELETE /api/customers/:customerId**
Delete a user's account
- **DELETE /api/reports/:customerId**
Delete a user's report
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The frontend will include input forms, dropdown menus, and buttons that allow users to submit requests to these API endpoints. Query results returned from the backend will be displayed in tables on the webpage

Acknowledgement of Using AI Tools

Yes. AI tools (e.g., ChatGPT) were used for language polishing, clarification of written explanations, word choice refinement, and document formatting. The timeline, task assignment and all technical content were independently produced by the project team.